Chen *et al. BMC Medical Education* (2025) 25:85
https://doi.org/10.1186/s12909-025-06705-x

BMC Medical Education

RESEARCH Open Access

Exploring effects of a booster workshop on progression and retention of resuscitation skills of residents when added to regular low-dose simulation

Hui-Wen Chen[1], Hung-Wei Tsai[2,3], Yi-Chun Chen[2,3], Yu-Jui Chiu[3], Enoch Yi-No Kang[4,5,6], Yi-Ting Hsu[3], S. Barry Issenberg[7*†] and Jen-Chieh Wu[3,8*†]

Abstract

Background Improving the resuscitation and teamwork skills of residents is key to better outcomes of in-hospital cardiac arrest events. This study aims to explore the effects of regular low-dose simulation combined with a booster workshop on the progression and retention of resuscitation skills and teamwork among residents.

Methods This comparative study took place at a teaching hospital in Northern Taiwan from August 2019 to June 2021. Residents were divided into two groups: the control group received regular low-dose simulation with faculty-led debriefing at the 1st, 6th, and 11th months, while the intervention group participated in an additional 3-hour simulation workshop one week before the 6th-month scenario. The workshop focused on resuscitation and teamwork skills. The study evaluated medical task performance, teamwork, patient safety attitudes, and timing of key resuscitation actions through standardized simulation scenarios, with the assessments occurring at the 1st, 6th, and 11th months (pre-test, mid-test, post-test, respectively).

Results Outcome measures in medical task performance and team leader behavior showed statistically significant improvement in both groups ($p < 0.05$) from pre-test to mid-test. After the intervention, the exposure group demonstrated significantly better results in medical task performance, team leader behavior, patient safety attitudes, and the timing of chest compression initiation compared to the control group ($p < 0.05$). Five months after the intervention, a significant decline in most measured outcomes was observed in the exposure group ($p < 0.05$). Despite this decline, the exposure group still performed significantly higher than the control group across most measured outcomes.

†S. Barry Issenberg and Jen-Chieh Wu these two authors contributed to the study equally.

*Correspondence:
S. Barry Issenberg
bissenbe@med.miami.edu
Jen-Chieh Wu
lendiswu@tmu.edu.tw

Full list of author information is available at the end of the article

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Conclusions This study demonstrates the potential and advantages of integrating simulation-based training into the clinical training curriculum for residents to improve medical task performance and teamwork behaviors for handling in-hospital cardiac arrest scenarios. The observed decline in skills over time supports the inclusion of regular refresher courses to maintain and advance these vital competencies.

Introduction

In-hospital cardiac arrest is a critical issue in health systems due to its association.

with high mortality and morbidity rates [1, 2]. Improvement in resuscitation quality is necessary within the healthcare system to ensure better outcomes for patients [3]. Adequate monitoring, timely implementation of essential interventions such as cardiopulmonary resuscitation, defibrillation and appropriate airway management, adherence to protocols, improvement in leadership and communication skills, such as teamwork behaviors, could improve outcomes [4, 5]. Training hospital clinical staffs to possess sufficient and efficient resuscitation and teamwork skills has been mandatory in many hospital systems for decades [6, 7]. Although various training modules have been developed [8], including simulation-based activities [9] and in-situ hands-on practice [10], it is important to investigate the effects of different training modules, particularly to assess the effectiveness and necessity of combining workplace teaching with additional simulation training.

In many academic training hospitals, resident physicians and primary healthcare providers form the initial response team. Until the resuscitation team arrives, these residents must manage and lead early resuscitation efforts during the critical first minutes of in-hospital cardiac arrests. Junior residents, particularly those in their first years of post-graduate training, often lack skills, experience, and confidence needed for effective clinical resuscitation [11]. Indeed, delays and errors occur by residents during cardiopulmonary arrest events [12].

In-hospital cardiac events demand immediate and efficient management, and delivering real-time feedback on residents' performance is challenging during resuscitation. Simulation training has been shown to have a positive impact on healthcare providers' resuscitation abilities and skills retention, and simulation assessment has a positive correlation with resuscitation competence in the workplace [13–15]. Previous evidence has shown that simulation can positively impact junior residents' resuscitation and leadership skills over time [16].

Research indicates that high-frequency low-dose in-situ simulation training can improve healthcare providers' resuscitation skills [17, 18]. It has been observed that more frequent training sessions, particularly those conducted at intervals of less than two months, are associated with improved outcomes [19]. The majority of these studies have utilized varying frequencies and primarily concentrated on resuscitation skills, rather than on teamwork behavior, which is a key factor of effective resuscitation. Studies have demonstrated that skills and knowledge decline over time, and booster or refresher courses are effective in maintaining performance levels [20]. Researchers have investigated various timings and frequencies for implementing such courses to help healthcare providers retain key skills; however, there is no definitive evidence indicating the most effective timing or frequency [21–23].

Understanding the learning outcomes of resuscitation and teamwork training for young residents is essential for designing effective programs that equip them with crucial skills. This study aimed to: (1) Explore the resuscitation abilities and teamwork behavior among residents with varying levels of resuscitation experience. (2) Explore the progression and retention of learning outcomes in resuscitation abilities and teamwork skills among residents following by regular low-dose simulation, with or without booster workshop. The alternative hypotheses were: [1] residents' resuscitation experience is associated with resuscitation abilities, and [2] there is a difference in resuscitation skills between residents who participate in a CPR booster workshop and those who only undergo regular low-dose simulation.

Methods

Study design and participants

This comparative study was conducted at a teaching hospital in Northern Taiwan, involving a cohort of residents between August 2019 and June 2021. The institutional review board approved this study of the Taipei Medical University (N202308050). Since August 2013, the medical education system in Taiwan underwent significant reform, shifting from a seven-year to a six-year curriculum. This reform resulted in an alteration in the training duration and structure for medical students. Specifically, students enrolled in the seven-year program had undergone two years of clerkship followed by one year of internship. In contrast, the revised six-year curriculum entailed only two years of clerkship, omitting the additional internship year. August 2019 marked the first time when graduates from six-year and seven-year medical programs simultaneously became residents. In this study, the research team identified residents who began their residencies in August 2019 as the regular low-dose group, while those starting in August 2020 were classified as the Simulation-Enhanced Team Cardiopulmonary

Resuscitation (SET-CPR) group since booster simulation course was launched at that time. The final cohort of students from the seven-year curriculum graduated in June 2019, resulting in a decline in the number of seven-year program graduates after 2020. This transition led to a significant disparity in the number of participants between the seven-year and six-year groups. The regular low-dose group consisted of 51 residents, including 21 from the six-year program and 30 from the seven-year program. The SET-CPR group included 38 residents including 27 from the six-year program and 11 from the seven-year program. All students in two groups underwent traditional clinical rotation training which consisted of three months in internal medicine, two months in surgical medicine, one month in obstetrics and gynecology, one month in pediatrics, one month in emergency medicine, two months in community medicine, and three months in electives of their choosing. Students in the regular low-dose group received standardized resuscitation simulation scenario exposure followed by faculty-led debriefing every 5 months (at the first, sixth, and eleventh months). These scenarios also served as pre-test, mid-test, and post-test assessments of performance. Students in the SET-CPR group, participated in an additional 3-hour workshop focused on resuscitation and teamwork training one week before the mid-test (6th month) to further enhance their skills (Fig. 1).

Educational theory implementation

Based on Kolb's Experiential Learning Theory [24], the simulation course was designed to engage residents in a hands-on and practical resuscitation scenario lasting 12 minutes. Under the instructor's guidance, students reflect on the scenario, recalling and analyzing events. This reflection enhances the understanding of relevant behaviors and teamwork dynamics. Building on reflective observations, individuals analyze the experience at an abstract level. This step, lasting 10 minutes, involves contextualizing events, forming theories, and understanding the broader principles of resuscitation protocols and teamwork behaviors. The residents engaged in another simulated scenario to manage the medical situation based on what they learned and planned. This learning cycle is repeated between 4 and 6 times during the leadership and resuscitation simulation training course.

Resuscitation and teamwork simulation-based training

The objectives of the resuscitation and teamwork simulation training workshop were to enable participants to efficiently apply resuscitation management and teamwork behavior in simulated In-hospital cardiac arrest scenario. The learning content was based on the TeamSTEPPS education module guidelines [25], and the American Heart Association's guidelines for cardiopulmonary resuscitation [7]. The simulation training content consisted of two parts: [1] resuscitation care patient management (assessment and management of the patient's airway, breathing, and circulation and the resuscitation of patients) [2], teamwork including leadership behavior (establishing members' roles and task responsibilities, guiding the team and communication, adherence and monitoring the care plan). Each small learning group consisted of seven to eight residents had multiple opportunities to practice at the simulation classroom. For each scenario, three to four residents were chosen to act as the primary response team for managing a simulated patient. One resident took the lead as the doctor, while two residents were assigned as nurses, fostering an environment that promoted leadership and teamwork skills. To maximum the learning of the participants, specific learning objectives and additional guidance were provided to the residents assigned the nurse roles. Following each scenario, the instructor facilitated a debriefing session with all participating residents, encouraging them to offer their insights and feedback, guided by the Gather, Analyze, Summarize (G.A.S.) debriefing model [26]. At the first, sixth and eleventh month, there is a twelve-minute simulation scenario followed by 10 min faculty-led debriefing session focused on the same content described above to serve as regular low-dose simulation scenario learning. The teaching team, consisting of a fifth-year emergency physician and two tenth-year emergency nurses, worked to ensure that all participants could equally practice leadership roles, engage actively in simulations, and contribute during debriefing sessions. This teaching team, which had over five years of experience in simulation-based instruction, provided comprehensive instruction to all residents, enriching their learning experience.

Outcome measurements

Assessment tools were employed to the management task performance and teamwork behavior performance by two trained raters. A questionnaire was designed to assess students' attitudes toward patient safety. The research team developed a standardized, twelve-minute simulation scenario and used the same scenario during the pre-test, mid-test, and post-test phases. Pre-tests were conducted in the first month of clinical rotation. Mid-tests were conducted at the end of the sixth month of the clinical rotation, which one week was after the SET-CPR group completed their simulation workshop. Post-test were conducted at the end of the eleventh month of the clinical rotation. The scenario involving two experienced practicing nurses acting as standardized nurses who interacted with the participant as embedded participants. The standardized nurses were instructed not to actively engage in executing procedures but rather

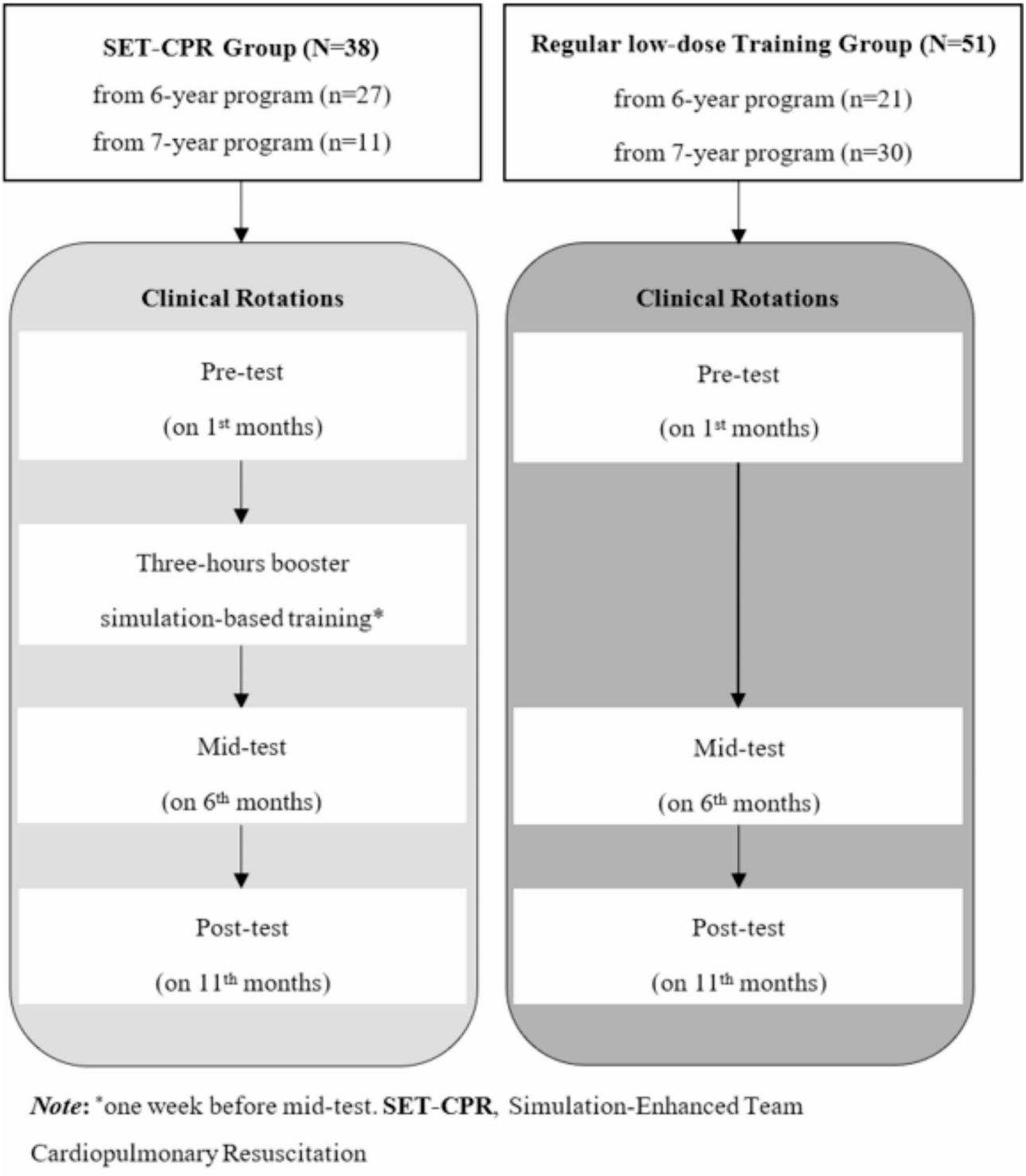

Fig.1. Flowchart of two groups

Fig. 1 Flowchart of two groups

to offer support to the trainees and deliver their orders. Ventricular fibrillation was programmed to occur at the seventh minute and continue until the end of the scenario. After each scenario assessment, there was a standardized 10-minute individual feedback conducted by trained faculty. To assess the impact of residents' prior resuscitation experiences on their learning outcomes, the research team compared the pre-test scenario scores between those with fewer than five resuscitation experiences and those with more than six.

Management task performance

The assessment of management task performance utilized a 54-item rating checklist [27]. The five main categories included: (1) initial assessment, (2) care environment setting evaluation, (3) airway management, (4) patient reevaluation, and (5) cardiac arrest management. Each item was scored on a dichotomous scale, assigning 1 for "Yes" and 0 for "No," resulting in a total score that ranged from 0 to 54. The higher the score, the better the management performance in a resuscitation situation. In addition, the total duration of the standardized assessment scenario was 12 min (720 s). The timing of two critical interventions was recorded: the moment when residents began chest compressions and the instance when the first defibrillation was administered upon the onset of ventricular fibrillation. If a resident did not perform cardiac compression or defibrillation, the time for these actions would be recorded as 300 s, reflecting the duration from the onset of ventricular fibrillation to the end of the scenario. The management task performance is categorized as nominal data, and Krippendorff's alpha was used assessing reliability with this type of data [28, 29]. According to Krippendorff's alpha, the interrater reliability was 0.89 for management task performance in the assessment scenarios, respectively [30].

Teamwork behavior performance

We assess the teamwork behavior performance according to the Mayo High Performance Teamwork Scale [31]. The researcher selected 9 items, including 8 essential items and 1 additional item related to the assessment scenario, from the original scale, ensuring their relevance and suitability for residents. Each item received a score ranging from 0 (never or rarely) to 2 (consistently), resulting in a total score ranging from 0 to 18. A higher score indicated better teamwork behavior performance. The teamwork behavior performance was measured using an ordinal scale, and the research team calculated Krippendorff's alpha for evaluating reliability with ordinal data [28, 29]. According to Krippendorff's alpha, the interrater reliability was 0.819 for team leader behavior performance in the assessment scenarios, respectively [30].

Patient safety attitude

The research team utilized a short-form version of safety attitude questionnaire which was based on one by Sexton [32]. The questionnaire consisted of 36 items divided into six dimensions with a 5-point Likert rating scale. The dimensions included: (1) Teamwork Climate, (2) Safety Climate, (3) Job Satisfaction, (4) Stress Recognition, (5) Perceptions of Management, and (6) Working Conditions. Scores for each item ranged from 1 (Strongly Disagree) to 5 (Strongly Agree), contributing to a cumulative score that fell within the range of 36 to 180. A higher score indicated a more positive attitude towards patient safety. The Scale reliability values of the questionnaire were 0.9.

Statistical analysis

The study presented descriptive statistics. Baseline demographic characteristics were analyzed using the chi-squared test. For data with non-normal distributions, nonparametric methods were applied, including the Mann–Whitney test for comparisons between groups and the Wilcoxon signed-rank test for within-group assessments. Repeated-measures analysis of variance was used to analyze data at three time points and explore changes within the group. All data analyses were conducted using IBM SPSS version 20, IBM Corp., Armonk, NY, USA. Statistical significance was determined using a two-tailed test, with a p-value < 0.05.

Results

In total, 89 residents were enrolled and completed in the training between August 2019 and June 2021. One data point related to the time to initiate chest compressions during the pre-test was missing in the regular low-dose group and was therefore excluded from the analysis. Statistically significant differences were found in the composition of groups between the 6-year and 7-year medical education systems ($p = 0.006$) (Table 1). In the SET-CPR group, there were 38 residents, with 27 from the 6-year program (71.1%) and 11 from the 7-year program (28.9%). In the regular low-dose group, there were 51 residents, with 21 from the 6-year program (41.2%) and 30 from the 7-year program (58.8%). The outcomes were further examined by stratifying them according to the medical education systems, acknowledging the disparities between the two groups and the importance of these variations within each system.

The performance and attitude scores of residents who participating fewer than 5 times and more than 6 times resuscitation events were shown in supplementary material 8. Seventy-seven residents participated in fewer than 5 resuscitation events, while 12 residents participated in more than 6. Significant higher patient safety scores were observed in group of residents with more than 6

Table 1 Demographic characteristic between groups ($N = 89$)

Variables	SET-CPR	Regular low-dose	*p*
	n (%)	*n* (%)	
Educational system			0.006*
6-year	27 (71.1)	21 (41.2)	
7-year	11 (28.9)	30 (58.8)	
Age (years), Mean (range)	26.53 (24–32)	26.55 (24–34)	0.57 [a]
Gender			0.07
Male	29 (76.3)	29 (56.9)	
Female	9 (23.7)	22 (43.1)	
Experiences of participating in the resuscitation of patients with cardiac arrest			0.09
None	2 (5.3)	11 (21.5)	
1–5	30 (78.9)	34 (66.7)	
6–10	5 (13.2)	6 (11.8)	
> 10	1 (2.6)	0 (0)	
The first personnel to discover a patient without pulse			1.00
Yes	2 (5.3)	2 (3.9)	
No	36 (94.7)	49 (96.1)	
Experience of initiating resuscitation			1.00
Yes	2 (5.3)	4 (7.8)	
No	36 (94.7)	47 (92.2)	

Note: Chi-Square Test; $^{*}p < 0.05$, two-tailed; [a] Mann-Whitney U test. SET-CPR, Simulation-Enhanced Team Cardiopulmonary Resuscitation. "The first personnel to discover a patient without a pulse" refers to the participant's experience as the first responder to identify a pulseless patient in a hospital setting. "Experience of initiating resuscitation" refers to the participant's involvement in activating the hospital's resuscitation response system when encountering a pulseless patient in need of immediate life-saving intervention

times participating resuscitation experience compared to those with less than 5 times (Median 153.75, Q1: 141.13, Q3: 168.75 vs. Median 141.00, Q1: 133.25, Q3: 151.00; $p = 0.01$). There were no statistically significant differences among other measured items.

The changes in resident performance and attitude scores within each group are shown in Fig. 2. From pre-test to mid-test, the scores for most outcome measures showed statistically significant improvement within the two groups ($p < 0.05$). The exceptions were patient safety attitude and time to initiate chest compression in regular low-dose training group. From mid-test to post-test, the scores for medical task performance, team behavior performance, patient safety attitude and time to deliver defibrillation showed statistically significant decline in SET-CPR group. In regular low-dose group, medical task performance, team behavior performance and time to initiate chest compression significantly improved from mid-test to post-test. From pre-test to post-test, both the SET-CPR and regular low-dose training groups demonstrated statistically significant progress in the overall learning trend ($p < 0.05$) except for patient safety attitude. This progress encompassed improvements in medical task performance and teamwork behavior, as well as reductions in the initial time for chest compression and time to first defibrillation. The details of data were showed in Table 2.

Residents' performance and attitude scores between groups are shown in Table 3. There are no statistically significant differences for most outcome measures at the pre-test between the two groups except for medical task performance score which is higher in regular low-dose training group ($p = 0.003$). At the mid-test, the SET-CPR training group had significantly higher scores than the regular low-dose training group for medical task performance, teamwork behavior performance, patient safety attitude, and time to initiation of chest compressions ($p < 0.05$). Specifically, at the post-test, the SET-CPR training group had significantly higher scores than the regular low-dose training group for medical task performance, and teamwork behavior performance ($p < 0.001$). Upon analyzing the subgroups within the educational system, results revealed that there are no statistically significant differences observed within the cohorts of 6-year or 7-year medical education system students across both groups (Supplementary Material 1 to 4). This suggests that the baseline difference was related to the composition between the groups.

Discussion

The findings in the present study show that resuscitation skills and patient safety attitude can improve significantly after initial training but decline over time. These are consistent with previous studies that report curricula integrated with simulation-based training have positive effects on healthcare providers' resuscitation knowledge and ability and patient outcomes [13, 15, 33], and that resuscitation knowledge and skills decline over time

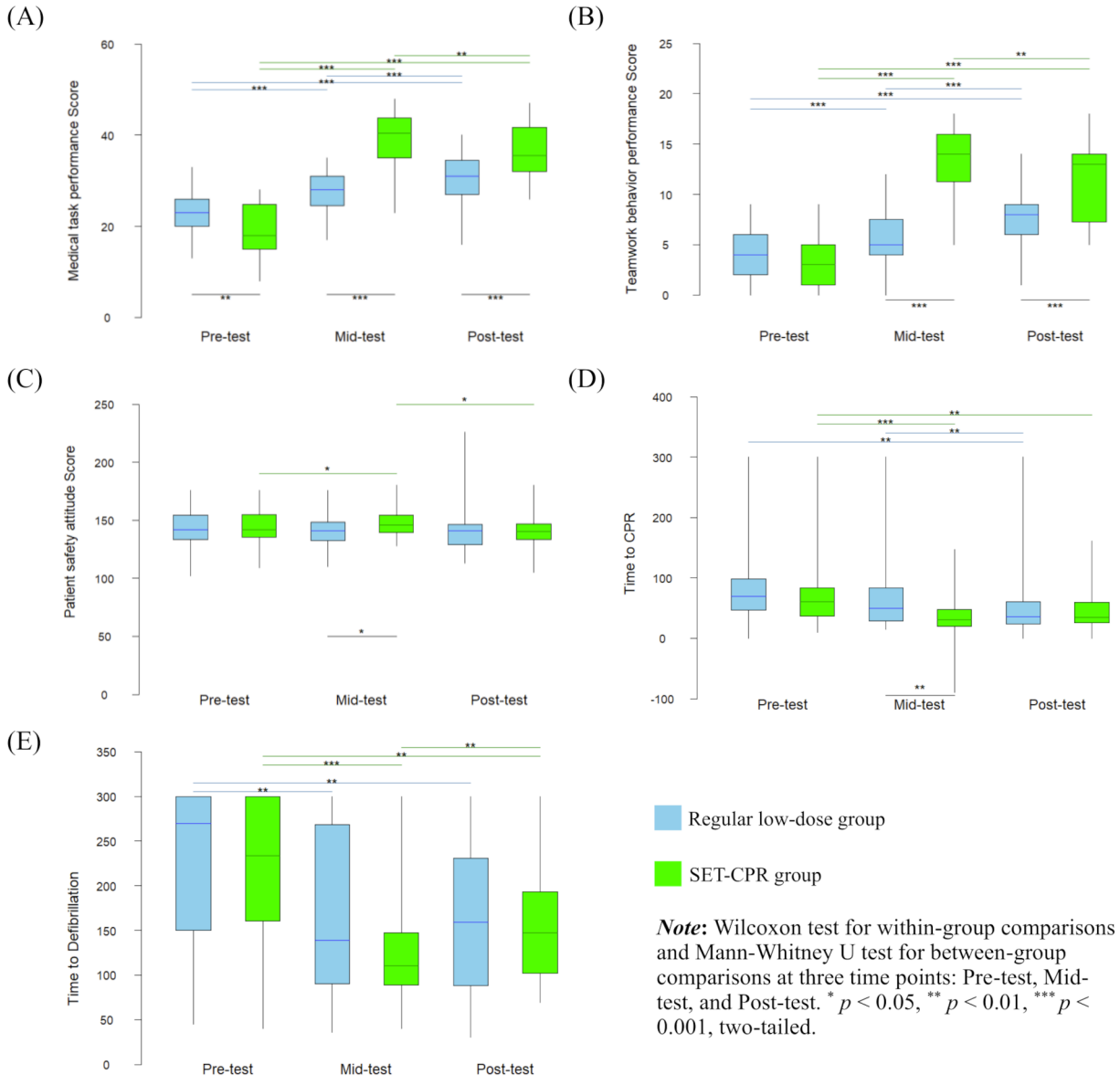

Fig. 2 A box plot is used to illustrate the learning outcomes between the two groups as well as the variations within each group. (**A**) Medical task performance, (**B**) Teamwork behavior performance, (**C**) Patient safety attitude, (**D**) Time to CPR, and (**E**) Time to defibrillation

[13, 34, 35]. Previous studies have also demonstrated that simulation-based training have a positive impact on development of leadership behavior in resuscitation team among residents and nurse practitioners [36–38]. These studies have not explored the decline in team and leadership capabilities, as well as patient safety attitude, following the initial resuscitation and team training course which were observed in the current study. Despite the challenges of simulating a profession within a simulation while maintaining focus on the overarching objectives [39], we noted enhancements in both medical performance and team dynamics, even among residents tasked with assuming the role of a nurse. This finding highlights the necessity for booster courses that not only focus on resuscitation skills but also on leadership, teamwork skills and patient safety attitude.

Most previous studies on booster training have focused on individual resuscitation skills [1, 23, 40–43]. Our findings are consistent with previous research indicating that booster training programs help sustain the skill performance levels of medical trainees, although the designs of these studies differ in terms of timing and frequency. Timing of booster training ranges from 12 h [43] to nine months [23] after primary training, and frequencies of

Table 2 Changes in medical task performance, teamwork behavior, and patient safety attitude within the two groups

Outcomes	Groups	*n*	Mid-test– Pre-test [a]		Post-test - Mid-test [a]		Post-test– Pre-test [a]		From Pre-test to Post-test [b]
			Median (Q1, Q3)	*p*	Median (Q1, Q3)	*p*	Median (Q1, Q3)	*p*	*p*
MTP (54 pts)									
	SET-CPR	38	18.50 (15.00, 25.25)	$<0.001^{***}$	-3.50 (-6.00, 1.00)	0.004^{**}	16.50 (13.00, 21.00)	$<0.001^{***}$	$<0.001^{***}$
	Regular low-dose	51	5.00 (1.00, 8.00)	$<0.001^{***}$	3.00 (-1.00, 6.00)	$<0.001^{***}$	7.00 (4.00, 10.00)	$<0.001^{***}$	$<0.001^{***}$
TBP (20 pts)									
	SET-CPR	38	10.00 (8.00, 12.25)	$<0.001^{***}$	-2.00 (-5.00, 1.25)	0.022^{**}	8.50 (5.00, 11.25)	$<0.001^{***}$	$<0.001^{***}$
	Regular low-dose	51	2.00 (0.00, 4.00)	$<0.001^{***}$	1.00 (0.00, 4.00)	$<0.001^{***}$	4.00 (2.00, 7.00)	$<0.001^{***}$	$<0.001^{***}$
PSA (180 pts)									
	SET-CPR	38	3.75 (-2.00, 11.63)	0.04^{*}	-2.75 (-17.13, 1.63)	0.03^{*}	-0.25 (-14.00, 6.25)	0.40	0.06
	Regular low-dose	51	-1.00 (-10.00, 7.00)	0.50	-1.00 (-4.00, 3.00)	0.50	-4.50 (-14.00, 7.00)	0.16	0.54
Time to CPR (sec.)									
	SET-CPR	38	-23.50 (-59.75, -7.75)	$<0.001^{***}$	0.50 (-18.50, 26.25)	0.56	-20.00 (-73.25, 5.25)	0.002^{**}	$<0.001^{***}$
	Regular low-dose	51	-21.50 (-56.25, 29.75)	0.13	-15.00 (-40.00, 8.00)	0.008^{**}	-24.00 (-75.75, 0.25)	0.003^{**}	0.01^{*}
Time to Defibrillation (sec.)									
	SET-CPR	38	-109.00 (-176.25, -37.50)	$<0.001^{***}$	37.50 (-5.50, 85.00)	0.005^{**}	-76.00 (-140.00, 0.00)	0.002^{**}	$<0.001^{***}$
	Regular low-dose	51	-30.00 (-170.00, 0.00)	0.001^{**}	0.00 (-54.00, 70.00)	0.58	-60.00 (-175.00, 10.00)	0.001^{**}	$<0.001^{***}$

Note: **MTP**, Medical task performance; **TBP**, Teamwork behavior performance; **PSA**, Patient safety attitude; **Time to CPR**, from cardiac arrest to initiate chest compression; **Time to Defibrillation**, from ventricular fibrillation attack to initiate defibrillation;

SET-CPR, Simulation-Enhanced Team Cardiopulmonary Resuscitation; **pts**, points. **sec**, seconds; $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, two-tailed; [a] = Wilcoxon test; [b] = Repeated measure analysis of variance

Table 3 Comparison of medical task performance, teamwork behavior, and patient safety attitudes between the two groups

Outcomes	Groups	n	Pre-test		Mid-test		Post-test	
			Median (Q1, Q3)	*p*	Median (Q1, Q3)	*p*	Median (Q1, Q3)	*p*
MTP (54 pts)				0.003^{**}		$<0.001^{***}$		$<0.001^{***}$
	SET-CPR	38	18.00 (15.00, 25.00)		40.50 (34.75, 44.00)		35.50 (32.00, 42.00)	
	Regular low-dose	51	23.00 (20.00, 26.00)		28.00 (24.00, 31.00)		31.00 (27.00,35.00)	
TBP (20 pts)				0.29		$<0.001^{***}$		$<0.001^{***}$
	SET-CPR	38	3.00 (1.00, 5.00)		14.00 (11.00,16.25)		13.00 (6.75, 14.25)	
	Regular low-dose	51	4.00 (2.00, 6.00)		5.00 (4.00, 8.00)		8.00 (6.00, 9.00)	
PSA (180 pts)				0.63		0.02^{*}		0.76
	SET-CPR	38	142.00 (134.63, 157.25)		145.75 (138.75, 155.00)		140.25 (133.00, 147.38)	
	Regular low-dose	51	141.50 (133.00, 157.00)		141.00 (132.00, 148.50)		141.00 (129.00, 147.00)	
Time to CPR (sec.)				0.27		0.006^{**}		0.86
	SET-CPR	38	60.50 (35.00, 91.75)		30.50 (20.00, 49.75)		35.00 (25.00, 61.25)	
	Regular low-dose	51	70.00 (45.75, 100.00)		50.00 (27.00, 85.00)		36.00 (23.00, 61.00)	
Time to defibrillation (sec.)				0.58		0.07		0.92
	SET-CPR	38	233.50 (157.50, 300.00)		110.50 (87.75, 150.25)		147.00 (100.00, 198.00)	
	Regular low-dose	51	270.00 (135.00, 300.00)		139.00 (85.00, 270.00)		159.00 (87.00, 235.00)	

Note: **MTP**, Medical task performance; **TBP**, Teamwork behavior performance; **PSA**, Patient safety attitude; **Time to CPR**, from cardiac arrest to initiate chest compression; **Time to Defibrillation**, from ventricular fibrillation attack to initiate defibrillation; **SET-CPR**, Simulation-Enhanced Team Cardiopulmonary Resuscitation; **pts**, points. **sec**, seconds; $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$, two-tailed; Mann - Whitney *U* test

booster trainings in the studies range from one session [23] to 11 sessions [44]. Future research should explore the optimal timing and frequency for booster training. However, the overall effectiveness of resuscitation extends beyond individual medical skills, as in-hospital resuscitation typically involves a collaborative team effort [4, 5]. Therefore, enhancing teamwork behaviors should be integrated into booster training. This study

contributes to the understanding of booster training for both CPR skills and teamwork in resuscitation.

Previous research shows that senior residents infrequently lead or participate in resuscitation scenario. And, while they were confident in performing many resuscitation skills, few performed critical components of these skills correctly [45]. In this study, there were no significant differences in resuscitation abilities and teamwork behavior behaviors between residents who had participated in fewer than five, or more than six resuscitation events. This suggests that for infrequent clinical events, additional learning activities are essential for junior medical residents to develop the necessary skills. Furthermore, it was found that the resuscitation abilities and teamwork behaviors of residents improved over time through conventional clinical training, which included exposure to regular low-dose simulation scenarios and faculty-led debriefings on their performance. This supports the inclusion of simulation activities and faculty-led debriefing sessions in teaching junior medical physicians how to handle rare, critical clinical events.

Moreover, when residents respond to a critical, clinical event it is challenging for clinical educators to observe the medical management and teamwork abilities of the residents. These circumstances challenge the assessment of residents' progress in resuscitation and teamwork abilities during cardiac arrest management. Simulation offers a standardized environment that enables clinical educators to assess the resuscitation skills, decision-making abilities, and teamwork behavior of residents. Simulation training also provides consistent learning opportunities for residents to develop the competency of rare yet crucial clinical events. As residents demonstrate mature capabilities in simulation assessments, clinical educators gain more opportunities to let residents directly handle patient care. This not only allows for direct observation of the residents' competencies but also ensures the maintenance of patient care quality.

For the simulated scenarios in the workshop of this study, the research team assigned one resident to act as the physician and two other residents to play the roles of nurses. This setup allowed participants to explore roles outside their professional background. Studies suggest that peer role-play enhances understanding of different roles and improves clinical skills [46, 47], while also reducing the demand for faculty resources. We note that assigning students to unrelated roles may reduce their attention to the learning objectives. However, previous evidence was from registered specialist nurses ($n = 9$; 64%) and midwives ($n = 1$; 7%) [39]. We advise that simulation educators should consider these factors alongside resource availability when designing simulation activities. Finally, research is warranted to explore the varying effects of peer role-play on targeted learning outcomes across different professions.

Limitations

This study has several limitations. First, the quasi-experimental design does not offer the same level of control over variables as a randomized controlled trial, which could potentially affect the generalizability of the findings. For instance, the medical education system serves as a pivotal determinant intricately intertwined with clinical experience, exhibiting notable disparities between the two groups under this study. Nonetheless, our analysis has systematically investigated the overarching patterns and differentials in outcomes across groups through the stratification of medical education systems. Second, the study was conducted at a single teaching hospital in Northern Taiwan, limiting the ability to generalize the results to other settings or healthcare systems with different training protocols and patient demographics. Third, residents were asked to assume the role of nurses during the simulation workshop scenarios. This change in role assignments may have impacted overall learning outcomes, as it shifted their focus from their usual responsibilities as physicians, potentially limiting the depth of their skill development in their primary role. While our booster training remained unchanged throughout the study period, the COVID-19 pandemic may have affected trainees' learning motivation or resuscitation skills. External factors such as disruptions to the learning environment, clinical workload, and changes in routine could have influenced trainees' motivation, potentially confounded the results and limited the ability to draw definitive conclusions about the SET-CPR's effectiveness.

Conclusion

This study shows that simulation-based training improves clinical performance and teamwork in cardiac arrest scenarios, but skills decline over time. Further research is needed to determine the best and most cost-effective ways to maintain these essential skills.

Abbreviations

SET-CPR	Simulation-Enhanced Team Cardiopulmonary Resuscitation
Team STEPPS	Team Strategies and Tools to Enhance Performance and Patient Safety

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12909-025-06705-x.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Supplementary Material 5

Supplementary Material 6

Supplementary Material 7

Supplementary Material 8

Supplementary Material 9

Acknowledgements
The research team would like to thank John M. O'Donnell for obtaining permission from the original development team to use the patient safety questionnaire in this study. We also extend our gratitude to the development team for granting access to this tool, which was integral to our research.

Author contributions
Chen HW, and Wu JC: writing– original draft, methodology. Chen HW, and Kang YN: software, formal analysis, data curation, visualization. Chen HW, Tsai HW, and Wu JC: conceptualization. Tsai HW, Chen YC, Chiu YJ, Hsu YT: investigation, resources, and data curation. Issenberg SB, and Wu JC (these two authors contributed to this article equally): review & editing, supervision, validation. All authors reviewed the manuscript.

Funding
This study did not receive any financial support.

Data availability
The datasets used and/or analyzed during the current study are intended for research publication. Researchers may contact the corresponding author to share data upon reasonable request.

Declarations

Ethics approval and consent to participate
The study was approved by the institutional review board approved this study of the Taipei Medical University (IRB # N202308050). After explaining the purposes and procedures of the study, informed consent was obtained from all students. All methods were conducted following the relevant guidelines and regulations.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Center for Education in Medical Simulation, Taipei Medical University, 250 Wuxing Street, Taipei 110301, Taiwan
[2]Department of Education and Humanities in Medicine, School of Medicine, College of Medicine, Taipei Medical University, 250 Wuxing Street, Taipei 110301, Taiwan
[3]Department of Emergency Medicine, Taipei Medical University Hospital, 252 Wuxing Street, Taipei 110301, Taiwan
[4]Institute of Health Policy & Management, College of Public Health, National Taiwan University, No. 1, Section 4, Roosevelt Rd, Taipei 106319, Taiwan
[5]Department of Health Care Management, College of Health Technology, National Taipei University of Nursing and Health Sciences, No. 365, Mingde Rd, Taipei 112303, Taiwan
[6]Evidence-Based Medicine Center, Wan Fang Hospital, Taipei Medical University, No. 111, Sec. 3, Xinglong Rd, Taipei 116079, Taiwan
[7]University of Miami Gordon Center for Simulation and Innovation in Medical Education, 1120 NW 14th St, Miami, FL 33136, USA
[8]Department of Education and Humanities in Medicine, School of Medicine, College of Medicine, Taipei Medical University, 250 Wuxing Street, Taipei 110301, Taiwan

Received: 6 June 2024 / Accepted: 13 January 2025
Published online: 17 January 2025

References

1. Andersen LW, Holmberg MJ, Berg KM, Donnino MW, Granfeldt A. In-Hospital Cardiac Arrest: Rev Jama. 2019;321(12):1200–10.
2. Nallamothu BK, Guetterman TC, Harrod M, Kellenberg JE, Lehrich JL, Kronick SL, et al. How do resuscitation teams at top-performing hospitals for In-Hospital cardiac arrest succeed? A qualitative study. Circulation. 2018;138(2):154–63.
3. Anderson ML, Nichol G, Dai D, Chan PS, Thomas L, Al-Khatib SM, et al. Association between hospital process composite performance and patient outcomes after In-Hospital cardiac arrest care. JAMA Cardiol. 2016;1(1):37–45.
4. Link MS, Berkow LC, Kudenchuk PJ, Halperin HR, Hess EP, Moitra VK, et al. Part 7: adult Advanced Cardiovascular Life support: 2015 American Heart Association guidelines Update for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation. 2015;132(18 Suppl 2):S444–64.
5. Yeung JH, Ong GJ, Davies RP, Gao F, Perkins GD. Factors affecting team leadership skills and their relationship with quality of cardiopulmonary resuscitation. Crit Care Med. 2012;40(9):2617–21.
6. Baldi E, Savastano S, Contri E, Lockey A, Conaghan P, Hulme J, et al. Mandatory cardiopulmonary resuscitation competencies for undergraduate healthcare students in Europe: a European Resuscitation Council guidance note. Eur J Anaesthesiol. 2020;37(10):839–41.
7. Cheng A, Magid DJ, Auerbach M, Bhanji F, Bigham BL, Blewer AL, et al. Part 6: Resuscitation Education Science: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation. 2020;142(16suppl2):S551–79.
8. Lauridsen KG, Løfgren B, Brogaard L, Paltved C, Hvidman L, Krogh K. Cardiopulmonary Resuscitation Training for Healthcare professionals: a scoping review. Simul Healthc. 2022;17(3):170–82.
9. McCoy CE, Rahman A, Rendon JC, Anderson CL, Langdorf MI, Lotfipour S, et al. Randomized Controlled Trial of Simulation vs. Standard Training for Teaching Medical Students High-quality cardiopulmonary resuscitation. West J Emerg Med. 2019;20(1):15–22.
10. Shrestha R, Badyal D, Shrestha AP, Shrestha A. In-situ Simulation-Based Module to Train interns in Resuscitation skills during Cardiac arrest. Adv Med Educ Pract. 2020;11:271–85.
11. Chandran KV, Abraham SV. Basic Life Support: need of the Hour-A study on the knowledge of Basic Life support among young doctors in India. Indian J Crit Care Med. 2020;24(5):332–5.
12. Hunt EA, Vera K, Diener-West M, Haggerty JA, Nelson KL, Shaffner DH, et al. Delays and errors in cardiopulmonary resuscitation and defibrillation by pediatric residents during simulated cardiopulmonary arrests. Resuscitation. 2009;80(7):819–25.
13. Au K, Lam D, Garg N, Chau A, Dzwonek A, Walker B, et al. Improving skills retention after advanced structured resuscitation training: a systematic review of randomized controlled trials. Resuscitation. 2019;138:284–96.
14. Weersink K, Hall AK, Rich J, Szulewski A, Dagnone JD. Simulation versus real-world performance: a direct comparison of emergency medicine resident resuscitation entrustment scoring. Adv Simul (Lond). 2019;4:9.
15. Young AK, Maniaci MJ, Simon LV, Lowman PE, McKenna RT, Thomas CS, et al. Use of a simulation-based advanced resuscitation training curriculum: impact on cardiopulmonary resuscitation quality and patient outcomes. J Intensive Care Soc. 2020;21(1):57–63.
16. Barsuk JH, Cohen ER, Wayne DB, Siddall VJ, McGaghie WC. Developing a Simulation-based mastery learning curriculum: lessons from 11 years of Advanced Cardiac Life Support. Simul Healthc. 2016;11(1):52–9.
17. Haynes J, Rettedal S, Perlman J, Ersdal H. A randomised controlled study of low-dose high-frequency In-Situ Simulation Training to Improve Newborn Resuscitation. Child (Basel). 2021;8(12).
18. Panchal AR, Norton G, Gibbons E, Buehler J, Kurz MC. Low dose- high frequency, case based psychomotor CPR training improves compression fraction for patients with in-hospital cardiac arrest. Resuscitation. 2020;146:26–31.
19. Haynes JC, Rettedal SI, Ushakova A, Perlman JM, Ersdal HL. How much training is Enough? Low-Dose, high-frequency Simulation Training and maintenance of competence in neonatal resuscitation. Simul Healthc. 2024.
20. Patel J, Posencheg M, Ades A. Proficiency and retention of neonatal resuscitation skills by pediatric residents. Pediatrics. 2012;130(3):515–21.
21. Boet S, Waldolf R, Bould C, Lam S, Burns JK, Moffett S, et al. Early or late booster for basic life support skill for laypeople: a simulation-based randomized controlled trial. Cjem. 2022;24(4):408–18.
22. Kerins J, McCully E, Stirling SA, Smith SE, Tiernan J, Tallentire VR. The impact of simulation-based mastery learning, booster session timing and clinical

exposure on confidence in intercostal drain insertion: a survey of internal medicine trainees in Scotland. BMC Med Educ. 2022;22(1):621.
23. Matterson HH, Szyld D, Green BR, Howell HB, Pusic MV, Mally PV, et al. Neonatal resuscitation experience curves: simulation based mastery learning booster sessions and skill decay patterns among pediatric residents. J Perinat Med. 2018;46(8):934–41.
24. Kolb DA, Boyatzis RE, Mainemelis C. Experiential learning theory: previous research and new directions. Perspectives on thinking, learning, and cognitive styles. Routledge; 2014. pp. 227–47.
25. Agency for Healthcare Research and Quality. TeamSTEPPS 2.0 Fundamentals. Rockville, MD; Updated Content last reviewed June 2019. Available from: https://www.ahrq.gov/teamstepps/instructor/fundamentals/index.html
26. Phrampus P, O'donnell J. Debriefing Using a Structured and Supported Approach. The comprehensive textbook of healthcare simulation. 2013:73–84.
27. Wu JC, Chen HW, Chiu YJ, Chen YC, Kang YN, Hsu YT, et al. Comparison of simulation-based interprofessional education and video-enhanced interprofessional education in improving the learning outcomes of medical and nursing students: a quasi-experimental study. Nurse Educ Today. 2022;118:105535.
28. Hughes J. Toward improved inference for Krippendorff's alpha agreement coefficient. J Stat Plann Inference. 2024;233:106170.
29. Marzi G, Balzano M, Marchiori D. K-Alpha calculator–Krippendorff's alpha calculator: a user-friendly tool for computing Krippendorff's alpha inter-rater reliability coefficient. MethodsX. 2024;12:102545.
30. Zapf A, Castell S, Morawietz L, Karch A. Measuring inter-rater reliability for nominal data - which coefficients and confidence intervals are appropriate? BMC Med Res Methodol. 2016;16:93.
31. Malec JF, Torsher LC, Dunn WF, Wiegmann DA, Arnold JJ, Brown DA, et al. The mayo high performance teamwork scale: reliability and validity for evaluating key crew resource management skills. Simul Healthc. 2007;2(1):4–10.
32. Sexton JB, Helmreich RL, Neilands TB, Rowan K, Vella K, Boyden J, et al. The Safety attitudes Questionnaire: psychometric properties, benchmarking data, and emerging research. BMC Health Serv Res. 2006;6:44.
33. Cheng A, Nadkarni VM, Mancini MB, Hunt EA, Sinz EH, Merchant RM, et al. Resuscitation Education Science: Educational Strategies to improve outcomes from Cardiac arrest: a Scientific Statement from the American Heart Association. Circulation. 2018;138(6):e82–122.
34. Greif R, Lockey A, Breckwoldt J, Carmona F, Conaghan P, Kuzovlev A, et al. [Education for resuscitation]. Notf Rett Med. 2021;24(4):750–72.
35. Partiprajak S, Thongpo P. Retention of basic life support knowledge, self-efficacy and chest compression performance in Thai undergraduate nursing students. Nurse Educ Pract. 2016;16(1):235–41.
36. Fernandez R, Rosenman ED, Olenick J, Misisco A, Brolliar SM, Chipman AK, et al. Simulation-based Team Leadership Training improves Team Leadership during actual trauma resuscitations: a Randomized Controlled Trial. Crit Care Med. 2020;48(1):73–82.
37. Roha YS, Lima EJ, Issenberg SB. Effects of an integrated simulation-based resuscitation skills training with clinical practicum on mastery learning and self-efficacy in nursing students. Collegian. 2016;23(1):53–9.
38. Ryan A, Rizwan R, Williams B, Benscoter A, Cooper DS, Iliopoulos I. Simulation Training improves Resuscitation Team Leadership skills of Nurse practitioners. J Pediatr Health Care. 2019;33(3):280–7.
39. Allvin R, Berndtzon M, Carlzon L, Edelbring S, Hult H, Hultin M, et al. Confident but not theoretically grounded - experienced simulation educators' perceptions of their own professional development. Adv Med Educ Pract. 2017;8:99–108.
40. Jeong HW, Moon SH. Enhancing ACLS training effects through video-assisted boosting methods: a comparative analysis of self-simulation and model simulation videos. Nurse Educ Pract. 2024;78:104010.
41. Sutton RM, Niles D, Meaney PA, Aplenc R, French B, Abella BS, et al. Booster training: evaluation of instructor-led bedside cardiopulmonary resuscitation skill training and automated corrective feedback to improve cardiopulmonary resuscitation compliance of Pediatric Basic Life Support providers during simulated cardiac arrest. Pediatr Crit Care Med. 2011;12(3):e116–21.
42. Sutton RM, Niles D, Meaney PA, Aplenc R, French B, Abella BS, et al. Low-dose, high-frequency CPR training improves skill retention of in-hospital pediatric providers. Pediatrics. 2011;128(1):e145–51.
43. Wolfe H, Maltese MR, Niles DE, Fischman E, Legkobitova V, Leffelman J, et al. Blood pressure Directed Booster Trainings improve Intensive Care Unit Provider Retention of excellent cardiopulmonary resuscitation skills. Pediatr Emerg Care. 2015;31(11):743–7.
44. Anderson R, Sebaldt A, Lin Y, Cheng A. Optimal training frequency for acquisition and retention of high-quality CPR skills: a randomized trial. Resuscitation. 2019;135:153–61.
45. Nadel FM, Lavelle JM, Fein JA, Giardino AP, Decker JM, Durbin DR. Assessing pediatric senior residents' training in resuscitation: fund of knowledge, technical skills, and perception of confidence. Pediatr Emerg Care. 2000;16(2):73–6.
46. Gelis A, Cervello S, Rey R, Llorca G, Lambert P, Franck N, et al. Peer role-play for Training Communication skills in Medical students: a systematic review. Simul Healthc. 2020;15(2):106–11.
47. Xu L, Liu W, Jiang X, Li Y. Impact of using peer role-playing on the clinical skills performance of pediatric trainees. BMC Med Educ. 2023;23(1):654.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.